

Then the average humid heat

$$c_s = 1.005 + 1.88 H \quad (33.3-6)$$

$$= 1.005 + 1.88(0.020) = 1.0426$$

$$\Rightarrow C_s = 1.0426 \times 10^3 \text{ J/kg} \cdot \text{K}$$

Calculating the bulk density of the bed ρ_s :

$$\epsilon = 1 - \rho_s / \rho \Rightarrow \rho_s = \rho(1 - \epsilon)$$

$$= (1522)(1 - 0.35) = 989.3$$

$$\Rightarrow \rho_{\text{bulk}} = 989.3 \text{ kg/m}^3$$

Finally, calculating the time required to dry the solid in the constant-rate period:

$$t = \frac{\rho_s \lambda_w x_1 (X_1 - X_c)}{G_{cs} (T_1 - T_w) (1 - e^{-\frac{h a x_1}{G_{cs}}})} =$$

$$= \frac{(989.3)(2637.45 \times 10^3)(0.0635)(0.90 - 0.45)}{(3220.0595)(1.0426 \times 10^3)(82.2 - 76.0) \left(1 - e^{-\frac{(107.7)(307)(0.0635)}{(3220)(1.0426 \times 10^3)}}\right)}$$

$$\Rightarrow t = 4806.948 \text{ sec} = 1.33526 \text{ hr.}$$

Koku
Adoboo

Homework #15

8/130

33-1

5/15

we know $T = 37.0^\circ\text{C}$ $P_{\text{atm}} = 101.3 \text{ kPa}$

Partial pressure of $\text{H}_2\text{O} = P_A = 3.59 \text{ kPa}$

Partial Pressure of Air $\Rightarrow P_B = P_{\text{atm}} - P_A$
 $= 101.3 - 3.59$

$$P_B = 97.71 \text{ kPa}$$

$$\text{Humidity} = \frac{\text{mass H}_2\text{O Vapor}}{\text{mass of Dry Air}} = \frac{m_A}{m_B}$$

$$\frac{P_A V}{P_B V} = \frac{\frac{m_A}{M_A} \times RT}{\frac{m_B}{M_B} \times RT} = \frac{P_A}{P_B} = \left(\frac{m_A}{m_B} \times \frac{M_B}{M_A} \right)$$

$$\frac{m_A}{m_B} = \frac{P_A}{P_B} \times \left(\frac{M_A}{M_B} \right)$$

$$H = \frac{m_A}{m_B} = P_A / P_B \times \left(\frac{M_A}{M_B} \right) = \frac{P_A}{P_B} \times \left(\frac{18.0152 \text{ g/mol}}{28.96 \text{ g/mol}} \right)$$

$$H = 0.622 \times \frac{P_A}{P_B} = 0.622 \times \frac{3.59 \text{ kPa}}{97.71 \text{ kPa}}$$

$$H = 0.02285 \frac{\text{kg H}_2\text{O}}{\text{kg dry Air}}$$

(1)

$$\textcircled{b} H_{\text{sat}} = 0.622 \left(\frac{P_{\text{sat}}}{P_{\text{tot}} - P_{\text{A sat}}} \right)$$

Vapor pressure from Antoine eq

$$\log (P_{\text{A}})_{\text{sat}} = A - \frac{B}{C + T}$$

$$\text{we know } A = 8.7131 \quad C = 233.426$$

$$B = 1730.63 \quad T \text{ is pure known}$$

$$\log (P_{\text{A}})_{\text{sat}} = 8.07181 - \left(\frac{1730.63}{233.426 + 37.8} \right)$$

$$(P_{\text{A}})_{\text{sat}} = (10)^{1.69}$$

$$= 49.039 \text{ mm Hg} = \frac{49.039}{760} \text{ atm}$$

$$= \frac{49.039}{760} \times (101.325 \text{ kPa})$$

$$P_{\text{sat}} = 6.538 \text{ kPa}$$

$$H_{\text{sat}} = 0.622 \times \left(\frac{6.538 \text{ kPa}}{101.3 - 6.538} \right)$$

$$H_{\text{sat}} = 0.0429 \frac{\text{kg H}_2\text{O}}{\text{kg dry Air}}$$

Percent Humidity

$$\% \text{ Humidity} = \frac{\text{humidity}}{\text{sat humidity}} \times 100$$

$$= \frac{H}{H_{\text{sat}}} \times 100 = \frac{0.02285}{0.0429} \times 100$$

$$= 53.26 \%$$

(2)

② % relative Humidity

$$\% \text{ Re Hum.} = \frac{P_A}{P_{\text{sat}}} \times 100$$

$$= \frac{3.59 \text{ kPa}}{6.538 \text{ kPa}} \times 100$$

$$= 54.91\% \approx 55\%$$

$$33.3 - 3 \rightarrow 15/15$$

• dew point 15.6°C at 100% Saturation
we got

$$H = 0.0113 \frac{\text{kg H}_2\text{O}}{\text{kg dry Air}}$$

from the Chart

• with $H = 0.0113 \frac{\text{kg H}_2\text{O}}{\text{kg dry Air}}$ &

$t = 65.6^\circ\text{C}$ (dry bulb temp, we have

$$H_p = 5.3\%$$

• Humidifying volume

$$V_H = (2.83 \times 10^3 + 4.56 \times 10^3 H) T$$

$$H = 0.0113 \text{ kg H}_2\text{O} / \text{kg dry Air}$$

$$T = 65.6^\circ\text{C} = 338.75 \text{ K}$$

$$③ V_H = (2.83 \times 10^3) + 4.56 \times 10^3 \times 0.0113 (338.75)$$

33-6-1

$$t_1 = \left(\frac{L_s}{AR} \right) \times (X_1 - X_2)$$

$$\text{New } L_s/A = \frac{24.4}{4} = 6.1$$

$$\text{Time of drying} = \left(\frac{6.1}{2.05} \right) \times (0.45 - 0.30)$$

$$t = 0.446 \text{ hrs}$$

16/20

6

356 - 3

20/2

we know $A = 0.61 \text{ m} \times 0.61 \text{ m}$
 $b = 25.4 \text{ mm}$
 $V = 3.05 \text{ m}^3$

$T = 60^\circ \text{C}$, $T_{wb} = 29.4^\circ \text{C}$

$X_1 = 0.035 \text{ kg H}_2\text{O/kg d.s}$

$X_2 = 0.22 \text{ kg H}_2\text{O/kg d.s}$

$L_3 = 11.34 \text{ kg d.s}$

$A + T = 60^\circ \text{C}$, $H = 0.0141 \text{ kg H}_2\text{O/kg d.s}$

$V_H = [(2.83 \times 10^3) + (4.56 \times 10^3) H] T$

$V_H = [2.83 \times 10^3 + (4.56 \times 10^3)(0.0141)](60 + 273)$

$V_H = 0.9638 \text{ m}^3 / \text{kg d.a}$

$m = 1.01 \text{ kg d.a} + 0.0141 \text{ kg H}_2\text{O}$

$m = 1.0141$

$\rho = \frac{m}{V_H} = \frac{1.0141}{0.9638} = 1.0522 \text{ kg/m}^3$

$G_1 = V \rho (3600) = (3.05 \times 3600) \text{ m}^3/\text{h} \times 1.0522 \text{ kg/m}^3$

$G_1 = 11553 \text{ kg/h-m}^2$

(7)

$$h = 0.0204 q^{0.08} \Rightarrow 0.0204 \times (11553)$$

$$h = 0.043 \frac{W}{m^2 \cdot K}$$

$$T_{wb} = 29.4^\circ C, h_w = 2432.1 \times 10^3$$

$$R_c = \frac{h}{h_w} (T - T_w) 3600$$

$$= \frac{36.29}{2432.14 \times 10^3} \times (60 - 29.4) 3600$$

$$R_c = 1.6437 \text{ kg/h} \cdot m^2$$

$$t = \frac{L_s}{A R_c} (x_1 - x_2) = \frac{11.34}{(0.61 \times 0.6)(1.6437)} \times (0.35 - 0.22)$$

$$t = 2.4103 \text{ hrs}$$

(5)

$$\frac{t_1}{t_2} = \frac{b_1}{b_2} \Rightarrow t_2 = \frac{t_1 b_2}{b_1} = \frac{2.4103 \text{ hr} \times 44.5 \text{ mm}}{25.4 \text{ mm}}$$

$$t_2 = 4.2228 \text{ hrs}$$

(8)

We know $33-10-3$

$\frac{20}{20}$

$$y_1 = 0.01 \text{ g H}_2\text{O / kg dry Air}$$

$$y = 0.031 \quad T_{\text{sat}} = 32^\circ\text{C}$$

$$G_s = 0.914$$

$$G = 0.914 + \frac{0.914 (0.031)}{0.01}$$

$$G = 0.923$$

$$E = 0.35 \text{ (void fraction)}$$

$$\text{Surface area of each particle} = \frac{\pi (0.0127)^2 (2)}{4 + \pi (0.0127)} = 2.508 \times 10^{-4} \text{ m}^2$$

$$dp = \left(\frac{2.508 \times 10^{-4}}{\pi} \right)^{0.5} = 8.935 \times 10^{-3} \text{ mm}$$

$$Re = \frac{dp G}{\mu} = \frac{8.935 \times 10^{-3}}{1.9 \times 10^{-5}} = 438.27$$

$$\mu = 1.9 \times 10^{-5}$$

$$N_{tG} = \frac{0.277}{(dp)^{0.35}} \left(\frac{dp}{\mu} \right)^{0.215} (x_{fs} z_s)^{0.64}$$

$$\ln \left[\frac{y_{\text{as}} - x_1}{y_{\text{as}} - x_2} \right] = N_{tG}$$

$$\ln \left[\frac{0.035 - 0.9}{0.031 - 0.45} \right] = 0.729 \times 100 = 72.5$$

$$J_s z_s = 1502 (0.0635) = 96.64$$

(10)